

Lab 4 — Flappy Bird Chat History

Project: Flappy Bird — Pygame Version

This document summarizes the work completed during the Lab-4 task. All text is black.

Task 1 — Refine Collision Detection

The collision issue occurred because the game checked only the bird's centre point. A pipe could overlap the bird's outer area while the centre remained outside, so the collision was missed.

Updated the collision checks in **game/game_engine.py** to use the bird's radius-based rectangle against both the top and bottom pipe rectangles. Existing gameplay, controls, pipe movement, and scoring were preserved.

Validation: edge-only overlaps were checked against both pipe rectangles, and the edited engine compiled successfully.

Task 2 — Game Over Screen

The engine already stopped updating after a collision, but the main loop only printed the final score. The window stayed open until it was closed.

Added a dimmed Game Over screen showing the final score and an exit prompt in **game/game_engine.py**. In **main.py**, a key press or mouse click exits after game over, and flap input is ignored once the game ends. Task 1 collision detection remained intact.

Validation: headless Pygame integration and Python compilation checks passed.

Task 3 — Replay and Difficulty Selection

Added Easy, Medium, and Hard difficulty choices to the Game Over screen, along with replay behavior. The screen also provides an Exit option. The user can select a difficulty by clicking a button or pressing keys 1–3; key 4 or Esc exits.

Validation: the game was run and the Game Over screen displayed the score and difficulty options.

Task 4 — Sound Feedback

Sound feedback was added, and the change was committed to the repository. The repository history shows the commit message "Add sound feedback."

Git and Repository Work

The project repository is: https://github.com/ChandanaRaniS/40_flappy-bird.git

Commits shown during the work include:

- Update README.md
- Fix pipe collision detection
- Add game over screen
- Add replay and difficulty selection
- Add sound feedback

Submission Checklist

- Before video showing the original behavior.

- After video showing the updated game and features.
- Updated code in the Lab-4 folder/repository.
- Chat history exported as a PDF.

Note: This is a concise project chat-history summary, not a verbatim transcript of every conversation message.